

Pneumatic End Effector Calculation Package

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Pin Selection & Spring Selection

The full calculations of both pin & spring sizing are outlined in the attached excel calculation. This section will outline the base case assumptions and calculation methodology used to evaluate the clamping force, spring feasibility, and pin size used for the end effector design. The following calculations outline the scaled down design parameters. The same calculations can be completed given regular-scale parameters, also provided in the excel sheet detailing calculations.

Clamping Force:

The following calculations outline the scaled down design parameters. The same calculations can be completed given regular-scale parameters, also provided in the excel sheet detailing calculations.

Assumptions:

- $m = 10\text{kg}$ (Maximum allowable mass based on allowable cylinder side loading)
- $g = 9.81\text{m/s}$
- $\mu_s = 0.8$ (Conservative estimate based on the friction coefficient between steel and rubber)

The clamping force (F_c) is evaluated via a force balance:

$$F_c = \frac{mg}{\mu_s} = \frac{(10)(9.81)}{0.8} = 122.5N$$

Spring Sizing:

To validate the spring size, based on the number of pins in action, the total deflection of each spring should not exceed the maximum compression displacement outlined in spring specifications.

Assumptions:

- $N = 6$ Pins in contact on each clamp (~40%) of pins in action at maximum clamping.

Selected Spring: 9657k5356 (McMaster-Carr) – 1.5” (38.1mm) Length Spring

- $k_s = 0.9632 \text{ kN/m}$, $l_{comp} = 15\text{mm}$

$$\begin{aligned}
 x_{allowable} &> x_{max} \\
 l - l_{comp} &> x_{max} \\
 l - l_{comp} &> \frac{Fc}{Nk_s} \\
 0.0381\text{m} - 0.015\text{m} &> \frac{122.5\text{N}}{6(963.2\text{N/m})} \\
 0.0381\text{m} - 0.015\text{m} &> \frac{122.5\text{N}}{6(963.2\text{N/m})} \\
 0.0232\text{m} &> 0.0212\text{m}
 \end{aligned}$$

Therefore, this spring is sufficient for rated loading conditions evaluated on base case assumptions:

Pin Sizing:

Assumptions:

- Worst case loading (Clamping force exerted on one pin)
- Cantilevered loading with point load at the edge of the pin (Largest bending & deflection)

Pin Material Properties & Geometry (Black-Oxide Alloy Steel):

- $E = 200 \text{ GPa}$
- $\sigma_y = 415 \text{ MPa}$
- $l = 1.5'' (0.0381\text{m})$
- $d = 1/4'' (0.00635\text{m})$

Determining the maximum deflection & factor of safety of the pin, we have:

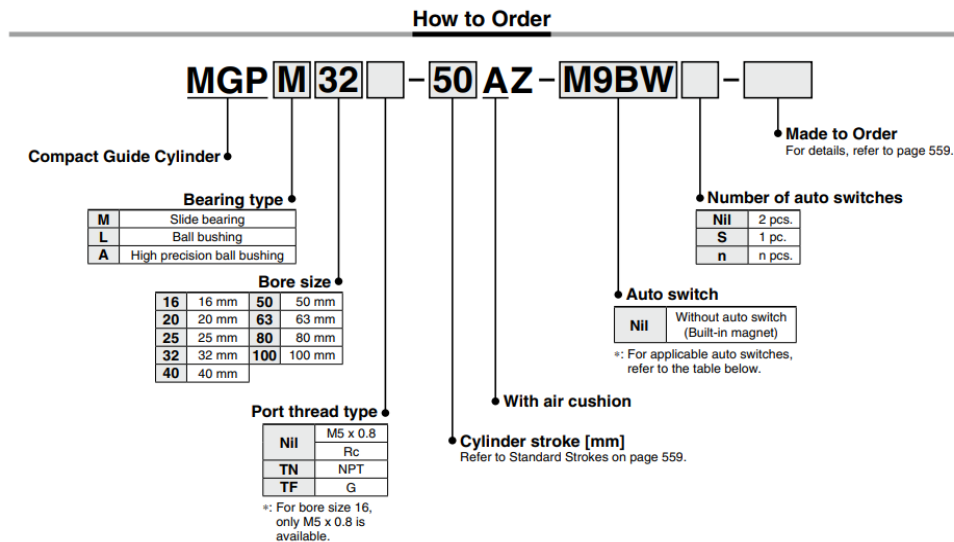
$$\begin{aligned}
 y &= -\frac{Fl^3}{3EI} = -\frac{mgl^3}{3E\frac{\pi}{4}r^4} = -\frac{64mgl^3}{3E\pi d^4} = -\frac{64(10)(9.81)(0.0381)^3}{3(200 \cdot 10^9)\pi(0.00635)^4} = -0.113\text{mm} \\
 &= \frac{(415 \cdot 10^6)\pi(0.00635)^3}{32(10)(9.81)(0.0381)} = 2.7
 \end{aligned}$$

Therefore, the maximum displacement and factor of safety of the pins under worst-case loading are within acceptable ranges.

Pneumatic Cylinder Selection & Fluid Power Calculations

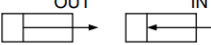
Cylinder Type: MGP Series – Compact Guide Cylinder

- Guided cylinders with 2 acting support rods to assist in experienced bending loads
- Must be designed to be within operating load capacities based on bore size



Theoretical Output – Desired Clamping Force

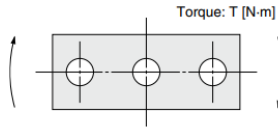
Theoretical Output



Bore size [mm]	Rod size [mm]	Operating direction	Piston area [mm²]	Operating pressure [MPa]								
				0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
12	6	OUT	113	23	34	45	57	68	79	90	102	113
		IN	85	17	25	34	42	51	59	68	76	85
16	8	OUT	201	40	60	80	101	121	141	161	181	201
		IN	151	30	45	60	75	90	106	121	136	151
20	10	OUT	314	63	94	126	157	188	220	251	283	314
		IN	236	47	71	94	118	141	165	188	212	236
25	10	OUT	491	98	147	196	245	295	344	393	442	491
		IN	412	82	124	165	206	247	289	330	371	412
32	14	OUT	804	161	241	322	402	483	563	643	724	804
		IN	650	130	195	260	325	390	455	520	585	650
40	14	OUT	1257	251	377	503	628	754	880	1005	1131	1257
		IN	1103	221	331	441	551	662	772	882	992	1103
50	18	OUT	1963	393	589	785	982	1178	1374	1571	1767	1963
		IN	1709	342	513	684	855	1025	1196	1367	1538	1709
63	18	OUT	3117	623	935	1247	1559	1870	2182	2494	2806	3117
		IN	2863	573	859	1145	1431	1718	2004	2290	2576	2863
80	22	OUT	5027	1005	1508	2011	2513	3016	3519	4021	4524	5027
		IN	4646	929	1394	1859	2323	2788	3252	3717	4182	4646
100	26	OUT	7854	1571	2356	3142	3927	4712	5498	6283	7069	7854
		IN	7323	1465	2197	2929	3662	4394	5126	5858	6591	7323

※: Theoretical output [N] = Pressure [MPa] x Piston area [mm²]

Allowable Rotational Torque of Plate



Bore size [mm]	Bearing type	Stroke [mm]																T [N-m]			
		10	20	25	30	40	50	75	100	125	150	175	200	250	300	350	400				
12	MGPM	0.39	0.32	—	0.27	0.24	0.21	0.43	0.36	0.31	0.27	0.24	0.22	0.19	—	—	—				
	MGPL/A	0.61	0.45	—	0.35	0.58	0.50	0.37	0.29	0.24	0.20	0.18	0.16	0.12	—	—	—				
16	MGPM	0.69	0.58	—	0.49	0.43	0.38	0.69	0.58	0.50	0.44	0.40	0.36	0.30	—	—	—				
	MGPL/A	0.99	0.74	—	0.59	0.99	0.86	0.65	0.52	0.43	0.37	0.32	0.28	0.23	—	—	—				
20	MGPM	—	1.05	—	0.93	0.83	0.75	1.88	1.63	1.44	1.28	1.16	1.06	0.90	0.78	0.69	0.62				
	MGPL/A	—	1.26	—	1.03	2.17	1.94	1.52	1.25	1.34	1.17	1.03	0.93	0.76	0.65	0.56	0.49				
25	MGPM	—	1.76	—	1.55	1.38	1.25	2.96	2.57	2.26	2.02	1.83	1.67	1.42	1.24	1.09	0.98				
	MGPL/A	—	2.11	—	1.75	3.37	3.02	2.38	1.97	2.05	1.78	1.58	1.41	1.16	0.98	0.85	0.74				
32	MGPM	—	—	6.35	—	—	5.13	5.69	4.97	4.42	3.98	3.61	3.31	2.84	2.48	2.20	1.98				
	MGPL/A	—	—	5.95	—	—	4.89	5.11	4.51	6.34	5.79	5.33	4.93	4.29	3.78	3.38	3.04				
40	MGPM	—	—	7.00	—	—	5.66	6.27	5.48	4.87	4.38	3.98	3.65	3.13	2.74	2.43	2.19				
	MGPL/A	—	—	6.55	—	—	5.39	5.62	4.96	6.98	6.38	5.87	5.43	4.72	4.16	3.71	3.35				
50	MGPM	—	—	13.0	—	—	10.8	12.0	10.6	9.50	8.60	7.86	7.24	6.24	5.49	4.90	4.43				
	MGPL/A	—	—	9.17	—	—	7.62	9.83	8.74	11.6	10.7	9.83	9.12	7.95	7.02	6.26	5.63				
63	MGPM	—	—	14.7	—	—	12.1	13.5	11.9	10.7	9.69	8.86	8.16	7.04	6.19	5.52	4.99				
	MGPL/A	—	—	10.2	—	—	8.48	11.0	9.74	13.0	11.9	11.0	10.2	8.84	7.80	6.94	6.24				
80	MGPM	—	—	21.9	—	—	18.6	22.9	20.5	18.6	17.0	15.6	14.5	12.6	11.2	10.0	9.11				
	MGPL/A	—	—	15.1	—	—	23.3	22.7	20.6	18.9	17.3	16.0	14.8	12.9	11.3	10.0	8.94				
100	MGPM	—	—	38.8	—	—	33.5	37.5	33.8	30.9	28.4	26.2	24.4	21.4	19.1	17.2	15.7				
	MGPL/A	—	—	27.1	—	—	30.6	37.9	34.6	31.8	29.3	27.2	25.3	22.1	19.5	17.3	15.5				

- In analysis, loading is assumed to be perfectly aligned with the center of actuation, meaning minimal to no plate torque is generated from operation
- In previous designs, eccentricity (>100mm) would have caused resultingly high plate torque making the design unsuitable for using even guided cylinders

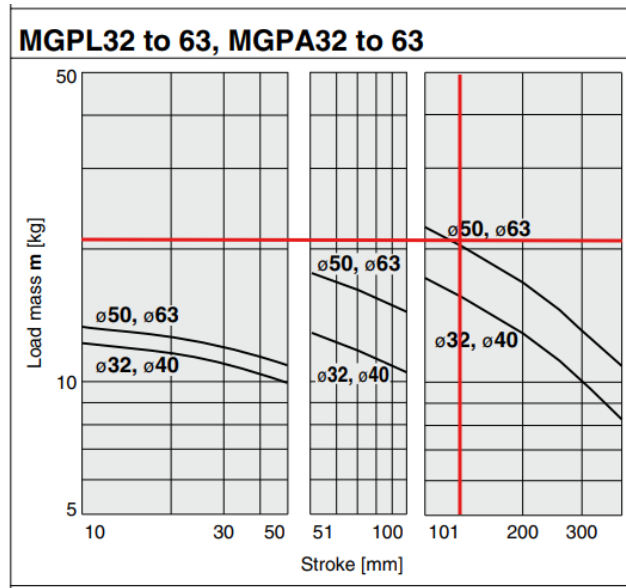
Selection Conditions

Mounting orientation	Vertical		Horizontal	
Maximum speed [mm/s]	200 or less	400	200 or less	400
Graph (Slide bearing)	(1), (2)	(3), (4)	(13), (14)	(15), (16)
Graph (Ball bushing)	(5) to (8)	(9) to (12)	(17), (18)	(19), (20)

- To determine if loading is appropriate given limitations to cylinder, horizontal loading under 200mm/s is required. Slow actuation is desirable to withstand inertial forces acting on the mechanism.

Regular Scale: MPGL-50-130Z

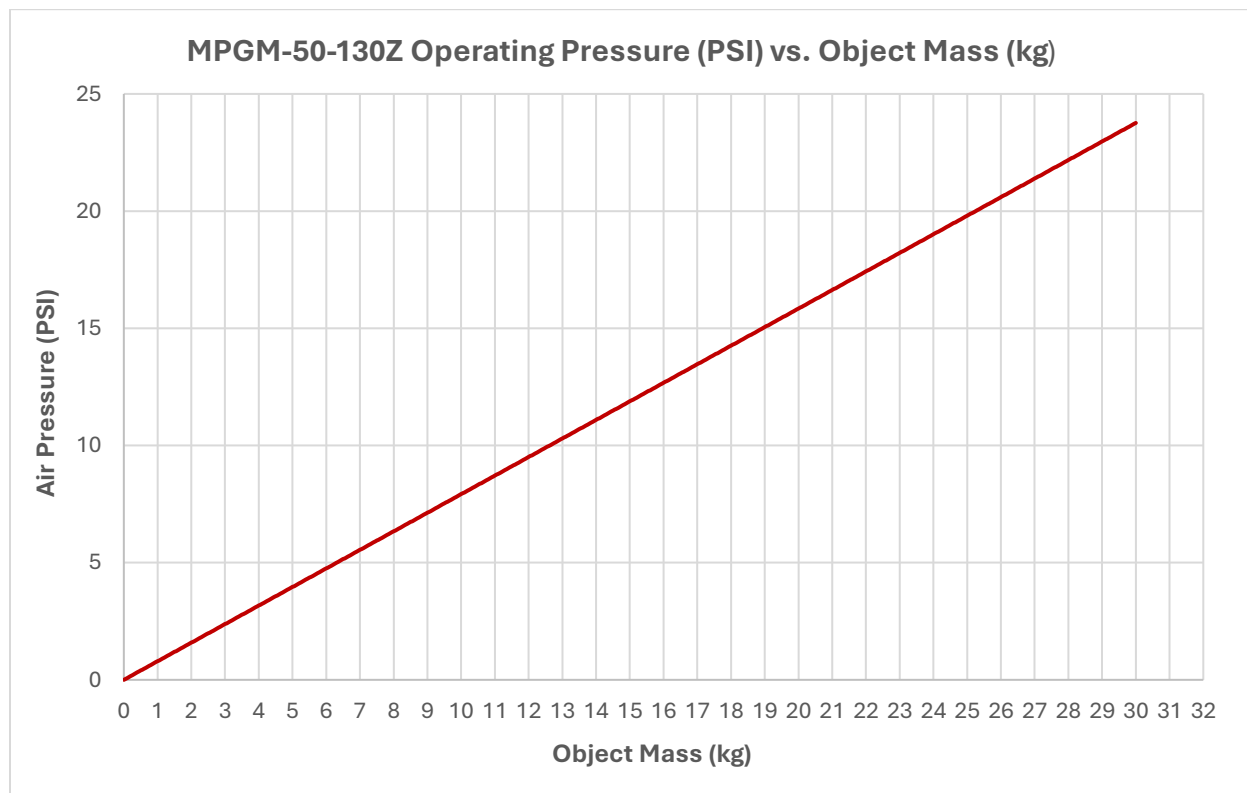
- Stroke length = 130mm, Bore Diameter = 50mm, Piston Diameter = 18mm
- Maximum Load Mass = 21kg
- For appropriate design, maximum allowable mass per arm = 20kg (-1kg)
- Max Operating Part Mass = $2 * (20 - 5) = 30\text{kg}$
- **Thus, the maximum desired part mass can be operated using MPGM-50-130Z.**



To determine the operating pressure for any mass, m , the following relationship is formed:

$$P = \frac{F_c}{A_{bore}} = \frac{145.038 \cdot \left(\frac{9.8m}{\mu}\right)}{10^6 \cdot \left(\frac{\pi}{4}\right) \cdot ((d_{bore})^2 - (d_{rod})^2)} = \frac{145.038 \cdot \left(\frac{9.8m}{0.8}\right)}{10^6 \cdot \left(\frac{\pi}{4}\right) \cdot ((0.5)^2 - (0.18)^2)} = 0.8m$$

$$P(\text{psi}) = 0.8 \cdot m \text{ (kg)}$$

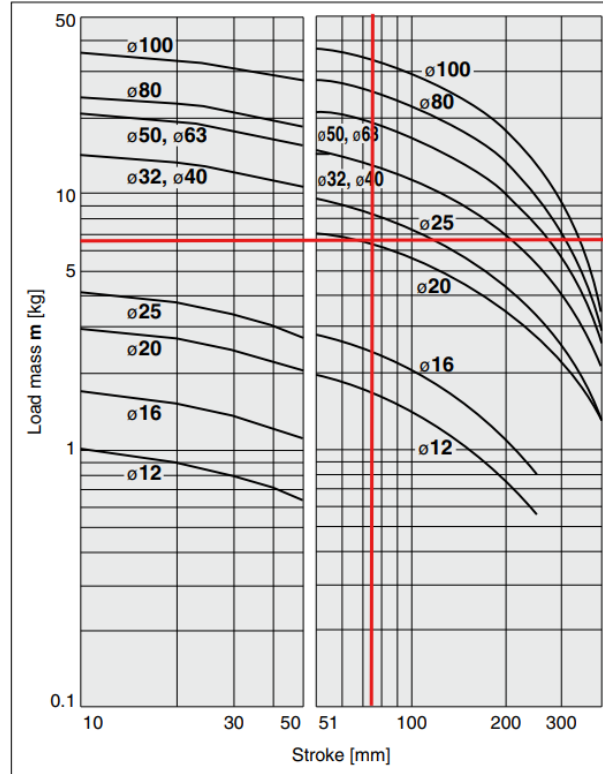


Scaled Down: MPGM-20-75Z

- Stroke length = 75mm, Bore Diameter = 20mm, Rod Diameter = 10mm
- Maximum Load Mass = 6.75kg
- For appropriate design, maximum allowable mass per arm = 5.75kg (-1kg)
- Max Operating Part Mass = $2 \cdot (5.75\text{kg} - \text{Mass of Arm}) = 2 \cdot (5.75 - 0.75) = 10\text{kg}$
- **Thus, the maximum desired part mass can be operated using MPGM-20-75Z.**

MGPM12 to 100

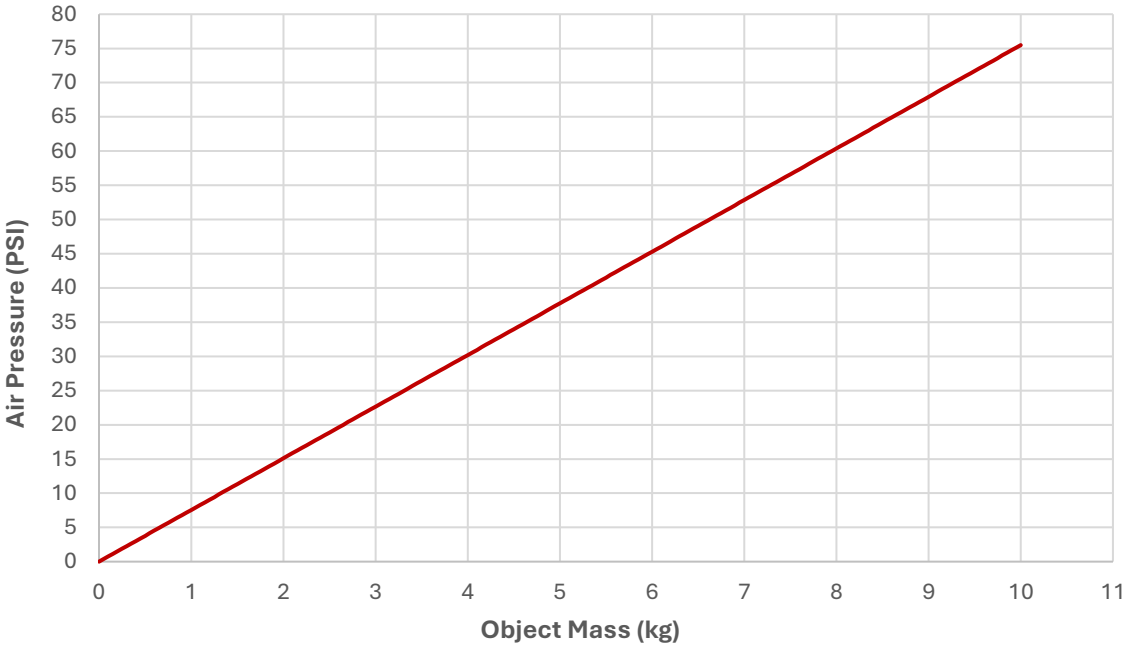
(13) L = 50 mm, V = 200 mm/s or less



$$P = \frac{F_c}{A_{bore}} = \frac{145.038 \cdot \left(\frac{9.8m}{\mu}\right)}{10^6 \cdot \left(\frac{\pi}{4}\right) \cdot ((d_{bore})^2 - (d_{rod})^2)} = \frac{145.038 \cdot \left(\frac{9.8m}{0.8}\right)}{10^6 \cdot \left(\frac{\pi}{4}\right) \cdot ((0.02)^2 - (0.01)^2)} = 7.54m$$

$$P(\text{psi}) = 7.54 \cdot m (\text{kg})$$

MPGM-20-75Z Operating Pressure (PSI) vs. Object Mass (kg)



Cylinder Clamp Connection Weldment Structural Analysis Results

- Material: Aluminum 6061 Alloy ($\sigma_Y = 280\text{MPa}$)
- Loading: Worst-case loading along centre of plate, 1.25x maximum load, bottom of frame
- Meshing: Generally lowest allowable element size before size limit is achieved

Preliminary Design: Regular Scale – Both Arms

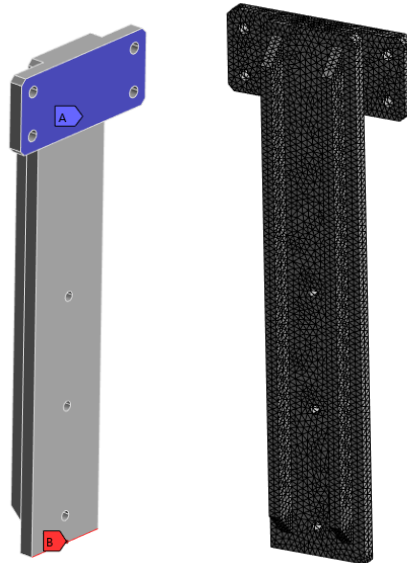
Stress Analysis Results	
Maximum Stress (MPa)	32 (SF = 8.5)
Max Deflection (mm)	1.2
Max Cycles (Zero Based, Goodman), Worst Case Loading (1.25Fc)	N = 10^8

Boundary/Loading Conditions, Meshing

- Assumed loading through center of plate, line pressure placed at bottom of plate for worst-case bending stress & max deflection; loading at rated clamping force (No force FOS applied)
- Fixed supports from back of cylinder plate
- Fillet welds not added, stress concentrations will result in higher stresses (worst-case)

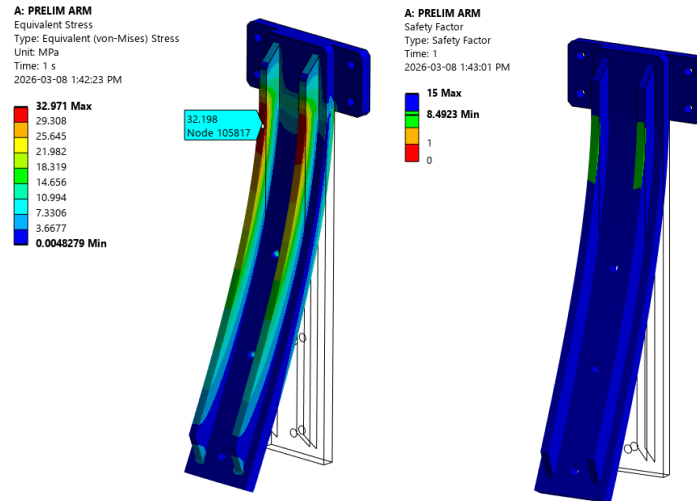
A: PRELIM ARM
Static Structural
Time: 1. s
2026-03-08 1:49:27 PM

A Fixed Support
B Force: 420. N

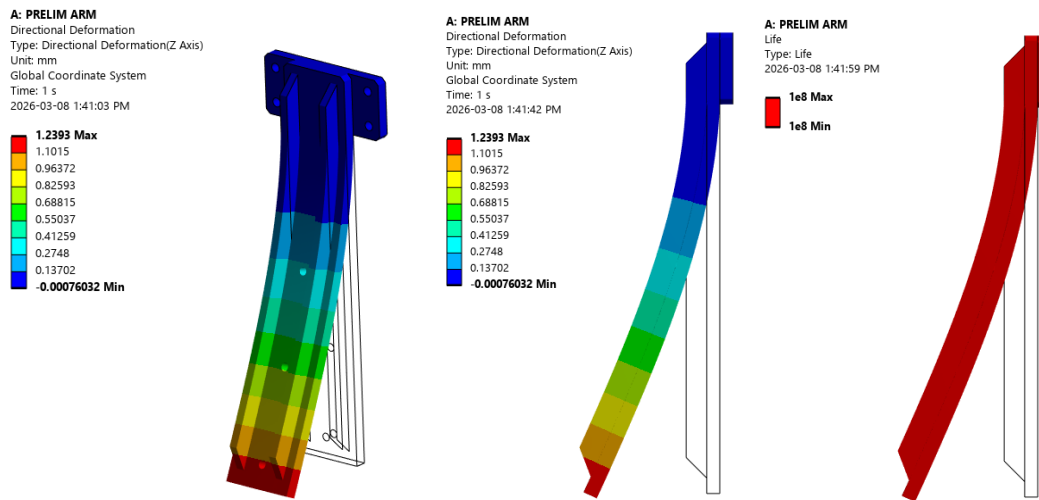


Details of "Body Sizing" - Sizing	
Scope	
Scoping Method	Geometry Selection
Geometry	1 Body
Definition	
Suppressed	No
Type	Element Size
☐ Element Size	5.0 mm
Advanced	
☐ Defeature Size	Default
Behavior	Soft

Equivalent (Von-Mises) Stress Results



Deflection & Cyclic Loading Results



V1: Regular Scale – Long Arm

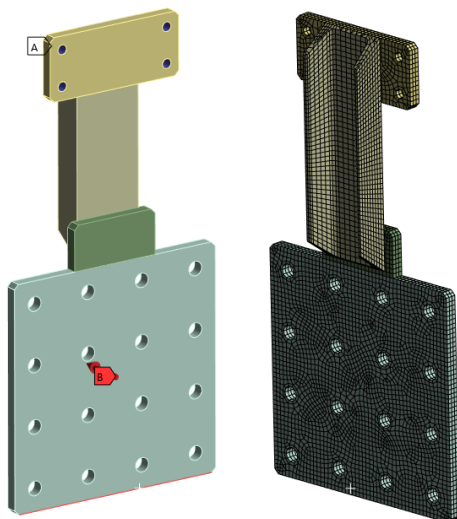
Stress Analysis Results	
Maximum Stress (MPa)	66 (SF = 4.2)
Max Deflection (mm)	2.01
Max Cycles (Zero Based, Goodman), Worst Case Loading (1.25Fc)	N=10 ⁸

Boundary/Loading Conditions, Meshing

- Assumed loading through center of plate, line pressure placed at bottom of plate for worst-case bending stress & max deflection; loading at ~1.25x maximum clamping force (~320N)
- Fixed supports from cylinder plate hole locations
- Fillet welds not added, stress concentrations will result in higher stresses (worst-case)

B: V1 ARM-FULL SCALE
 Static Structural
 Time: 1 s
 2026-03-08 1:26:51 PM

A Fixed Support
B Remote Force: -400. N

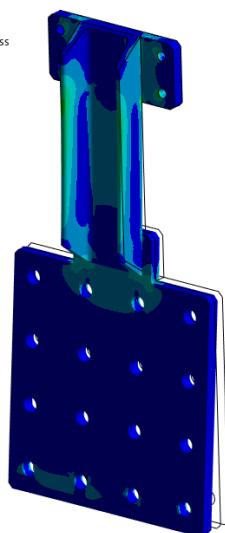


Details of "Body Sizing" - Sizing	
Scope	
Scoping Method	Geometry Selection
Geometry	4 Bodies
Definition	
Suppressed	No
Type	Element Size
☐ Element Size	5.0 mm
Advanced	
☐ Defeature Size	Default
Behavior	Soft

Equivalent (Von-Mises) Stress Results

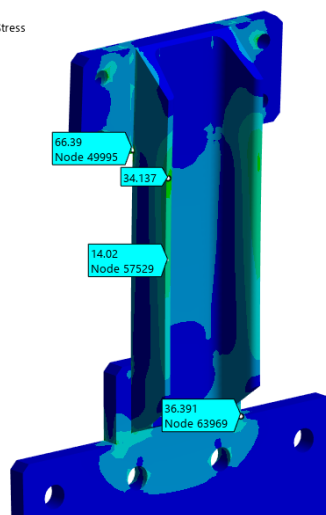
B: V1 ARM-FULL SCALE
 Equivalent Stress
 Type: Equivalent (von-Mises) Stress
 Unit: MPa
 Time: 1 s
 2026-03-08 1:27:21 PM

67.158
66.39 Max
 58.766
 50.374
 41.982
 33.59
 25.199
 16.807
 8.4149
0.023101 Min

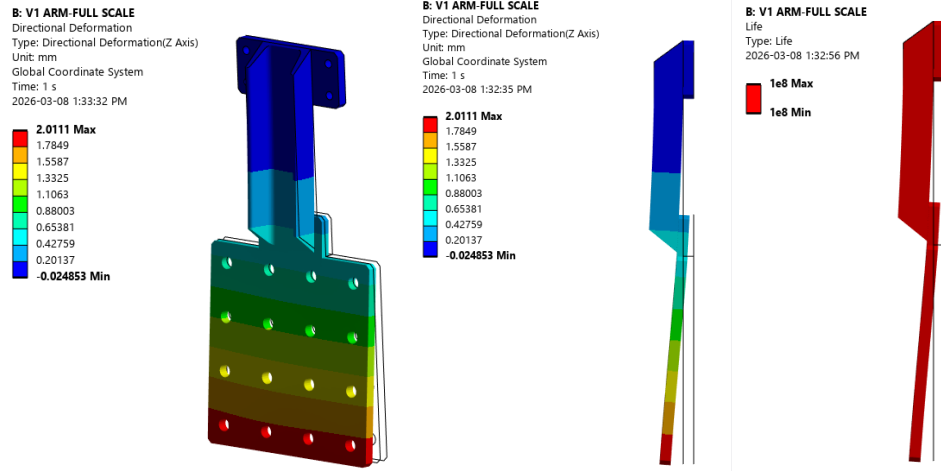


B: V1 ARM-FULL SCALE
 Equivalent Stress
 Type: Equivalent (von-Mises) Stress
 Unit: MPa
 Time: 1 s
 2026-03-08 1:29:47 PM

67.158
66.39 Max
 58.766
 50.374
 41.982
 33.59
 25.199
 16.807
 8.4149
0.023101 Min



Deflection & Cyclic Loading Results



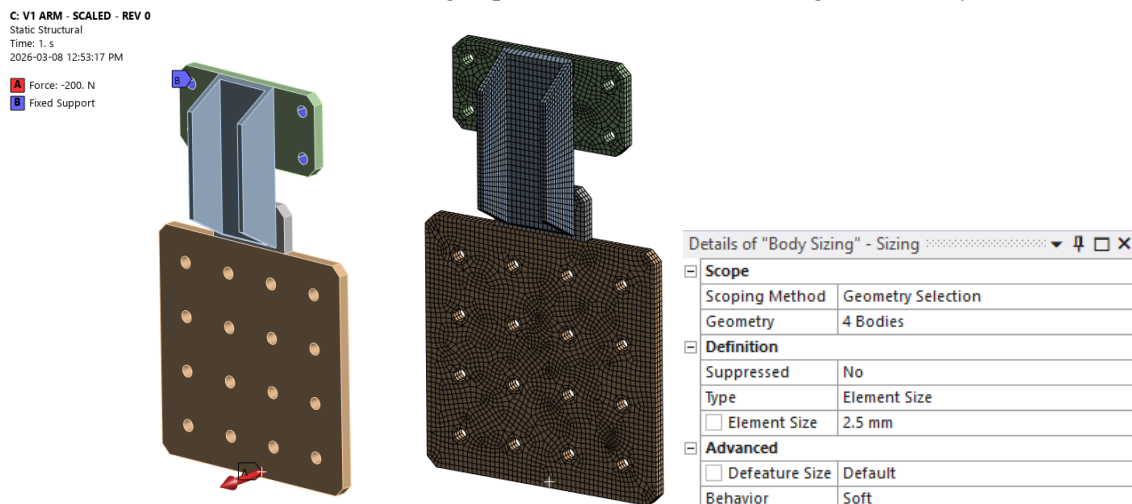
V1: Scaled Down

Dynamic Weldment

Stress Analysis Results	
Maximum Stress (MPa)	184 (SF = 1.52)
Max Deflection (mm)	2.05
Max Cycles (Zero Based, Goodman), Worst Case Loading (1.5Fc)	N=1,167,200

Boundary/Loading Conditions, Meshing

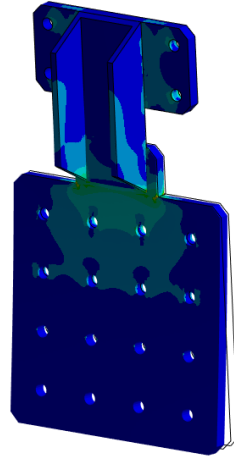
- Assumed loading through center of plate, line pressure placed at bottom of plate for worst-case bending stress & max deflection; loading at ~1.5x maximum clamping force (~125N)
- Fixed supports from cylinder plate hole locations
- Does not contain a fillet, increasing expected stresses and reducing available cycles



Equivalent (Von-Mises) Stress Results

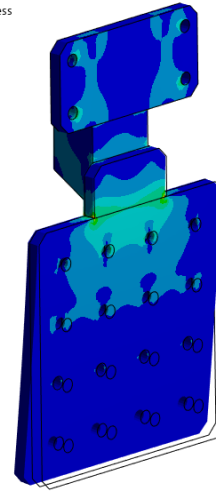
C: V1 ARM - SCALED - REV 0
 Equivalent Stress
 Type: Equivalent (von-Mises) Stress
 Unit: MPa
 Time: 1 s
 2026-03-08 12:55:00 PM

183.96 Max
 163.53
 143.1
 122.66
 102.23
 81.801
 61.37
 40.939
 20.508
 0.076344 Min



C: V1 ARM - SCALED - REV 0
 Equivalent Stress
 Type: Equivalent (von-Mises) Stress
 Unit: MPa
 Time: 1 s
 2026-03-08 12:57:44 PM

183.96 Max
 163.53
 143.1
 122.66
 102.23
 81.801
 61.37
 40.939
 20.508
 0.076344 Min



Deflection & Cyclic Loading Results

C: V1 ARM - SCALED - REV 0
 Directional Deformation
 Type: Directional Deformation(X Axis)
 Unit: mm
 Global Coordinate System
 Time: 1 s
 2026-03-08 1:00:46 PM

0.034395 Max
 -0.1956
 -0.42559
 -0.65558
 -0.88557
 -1.1156
 -1.3456
 -1.5755
 -1.8055
 -2.0355 Min



C: V1 ARM - SCALED - REV 0
 Life
 Type: Life
 2026-04-03 2:20:44 PM

1e8 Max
 6.0987e7
 3.7194e7
 2.2684e7
 1.3834e7
 8.437e6
 5.1455e6
 3.1381e6
 1.9138e6
 1.1672e6 Min



Static Weldment

- Fillet weld added on clamp plate connection piece to lower stress concentration
- Angle thickness increased from 0.25 to 0.5in to reduce bending stress on component

Stress Analysis Results	
Maximum Stress (MPa)	137 (SF = 2.05)
Max Deflection (mm)	2.85
Max Cycles (Zero Based, Goodman), Worst Case Loading (1.5Fc)	N = 65,686,000

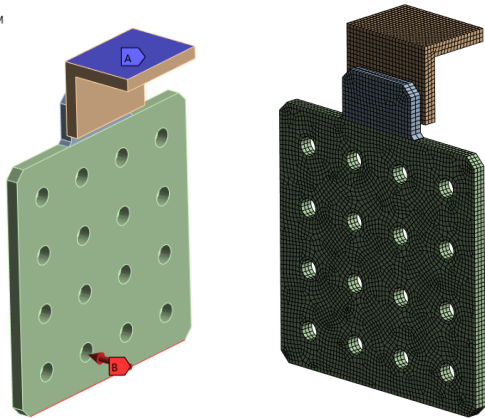
Boundary/Loading Conditions, Meshing

- Assumed loading through center of plate, line pressure placed at bottom of plate for worst-case bending stress & max deflection; loading at ~1.5x maximum clamping force (~125N)

D: Copy of V1 ARM - SCALED - REV 1 - FILLET WELD ADDED

Static Structural
Time: 1 s
2026-03-08 12:35:46 PM

A Fixed Support
B Force: 200. N



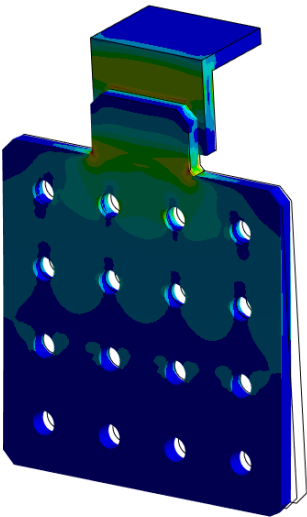
Details of "Face Sizing" - Sizing	
[-] Scope	
Scoping Method	Geometry Selection
Geometry	44 Faces
[-] Definition	
Suppressed	No
Type	Element Size
☐ Element Size	2.0 mm
[-] Advanced	
☐ Defeature Size	Default (5.4491e-002 mm)
Influence Volume	No
Behavior	Soft
☐ Growth Rate	Default (1.85)
Capture Curvature	No
Capture Proximity	No

Equivalent (Von-Mises) Stress Results

D: Copy of V1 ARM - SCALED - REV 1 - FILLET WELD ADDED

Equivalent Stress
Type: Equivalent (von-Mises) Stress
Unit: MPa
Time: 1 s
2026-03-07 4:04:43 PM

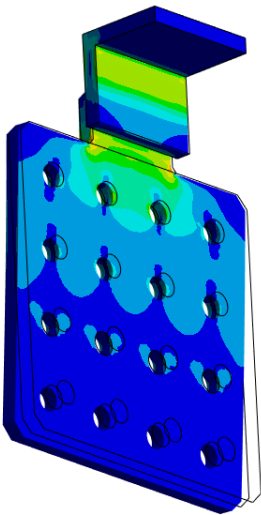
136.58 Max
121.4
106.23
91.054
75.879
60.704
45.529
30.354
15.179
0.0036622 Min



D: Copy of V1 ARM - SCALED - REV 1 - FILLET WELD ADDED

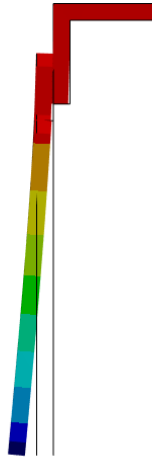
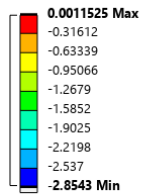
Equivalent Stress
Type: Equivalent (von-Mises) Stress
Unit: MPa
Time: 1 s
2026-03-07 4:04:43 PM

136.58 Max
121.4
106.23
91.054
75.879
60.704
45.529
30.354
15.179
0.0036622 Min

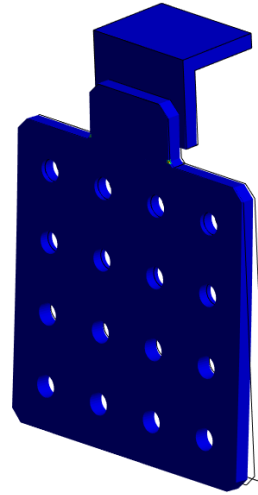
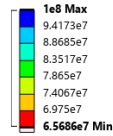


Deflection & Cyclic Loading Results

D: Copy of V1 ARM - SCALED - REV 1 - FILLET WELD ADDED
 Directional Deformation
 Type: Directional Deformation(X Axis)
 Unit: mm
 Global Coordinate System
 Time: 1 s
 2026-03-07 4:15:54 PM



D: Copy of V1 ARM - SCALED - REV 1 - FILLET WELD ADDED
 Life
 Type: Life
 2026-04-03 2:22:01 PM



Structural Analysis Results (ANSYS) Summary

	Preliminary Design	Refined Design Long Arm	Refined Design (Scaled) Static	Refined Design (Scaled) Dynamic
Loading Condition	1.25Fc	1.25Fc	1.5Fc	1.5Fc
Maximum Stress (MPa)	32	66	137	184
Safety Factor (SF)	8.5	4.2	2.05	1.52
Max Deflection (mm)	1.2	2.01	2.85	2.05
Max Cycles (Zero Based, Goodman), Worst Case Loading (1.5Fc)	N >10e8	N >10e8	N = 65,686,000	N=1,167,200